Linear Algebra — Lecture 5

Change of Coordinate Matrix; Dual Spaces; the Rank of a Matrix

Friedberg–Insel–Spence, *Linear Algebra* 4e, §2.5, §2.6, §3.1–§3.2
D. C. Lay, *Linear Algebra and Its Applications* 4e, §2.2–§2.3, §4.4, §4.6–§4.7

Topics. Coordinates and the coordinate mapping, the change of coordinate matrix, similar matrices, dual spaces and the transpose, and the rank of a matrix.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Coordinates, mapping	§4.4 (p. 216–221)	§2.5 (p. 110–111)
Change of coordinate matrix	§4.7 (p. 239–242)	§2.5 (p. 111–112)
Similar matrices	—	§2.5 (p. 112–115)
Linear functionals	—	§2.6 (p. 119–121)
Dual basis, transpose	—	§2.6 (p. 120–122)
The double dual	—	§2.6 (p. 121–123)
Elementary matrices	§2.2 (p. 102–108)	§3.1 (p. 148–150)
The rank of a matrix	§4.6 (p. 230–236)	§3.2 (p. 152–159)

Two questions that drive the whole lecture: (1) *After a change of basis, how do the coordinates of a vector and the matrix of an operator change?* (2) *How does the number "the dimension of the image of a matrix" behave under a change of basis, under elementary operations and under multiplication?*

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. Lay: PDF page = printed page +17. Friedberg: PDF page = printed page +15.

1 Coordinates and the Change of Coordinate Matrix

Lecture 3 turned an ordered basis $\beta = \{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ of V into a coordinate machine: every $\boldsymbol{x} \in V$ is written uniquely as $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$, and $[\boldsymbol{x}]_\beta$ is that coefficient vector. This first section does one thing: it **changes the machine** and asks how the coordinates and the matrices move with it.

Definition 1.1 (Coordinate vector; Lay §4.4, p. 216)**.** Let $\beta = \{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ be an ordered basis of V. For $\boldsymbol{x} \in V$, the scalars $c_1, \dots, c_n$ in $\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$ are the **coordinates** of $\boldsymbol{x}$ relative to β, the column

$$[\boldsymbol{x}]_\beta = (c_1, \dots, c_n)^\top$$

is the **coordinate vector** of $\boldsymbol{x}$, and $\boldsymbol{x} \mapsto [\boldsymbol{x}]_\beta$ is the **coordinate mapping**.

Theorem 1.1 (Unique Representation Theorem; Lay Theorem 7, §4.4, p. 216)**.** *Let β be a basis of V. Then every $\boldsymbol{x} \in V$ has exactly one expansion*

$$\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$$

in the vectors of β.

Proof. Existence is "β spans V"; uniqueness is "β is linearly independent": if $\sum_i c_i \boldsymbol{b}_i = \sum_i d_i \boldsymbol{b}_i$, then $\sum_i (c_i - d_i)\boldsymbol{b}_i = \boldsymbol{0}$, so all $c_i - d_i = 0$. □

Theorem 1.2 (The coordinate mapping is an isomorphism; Lay Theorem 8, §4.4, p. 219)**.** *Let β be an ordered basis of an n-dimensional space V. Then $\boldsymbol{x} \mapsto [\boldsymbol{x}]_\beta$ is an **isomorphism** from V onto $\mathbb{F}^n$.*

Proof. Linearity: coordinates are determined by the basis, so if $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$ and $\boldsymbol{y} = \sum_i d_i \boldsymbol{b}_i$, then $\boldsymbol{x} + \boldsymbol{y} = \sum_i (c_i + d_i)\boldsymbol{b}_i$, that is, coordinates add componentwise; the same holds for scalar multiples. Injectivity: if $[\boldsymbol{x}]_\beta = \boldsymbol{0}$, then $\boldsymbol{x} = \sum_i 0 \cdot \boldsymbol{b}_i = \boldsymbol{0}$. Surjectivity: given $(c_1, \ldots, c_n)^\top$, take $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$. □

Definition 1.2 (Change of coordinate matrix; Friedberg §2.5, p. 110–111)**.** Let β, β' be two ordered bases of V. The matrix

$$Q = [I_V]^{\beta}_{\beta'}$$

is the **change of coordinate matrix** that changes β'-coordinates into β-coordinates; by the definition of a matrix representation, its **j-th column** is the coordinate vector $[\boldsymbol{b}'_j]_\beta$ of the j-th vector of β'.

Theorem 1.3 (How coordinates change; Friedberg Theorem 2.22, §2.5, p. 111)**.**

(a) Q is invertible;
(b) for every $\boldsymbol{v} \in V$, $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$.

Proof. (a) I_V is invertible, and "invertible if and only if its matrix representation is invertible" (Lecture 4 notes, Theorem 4.3; Friedberg Theorem 2.18) makes $Q = [I_V]^{\beta}_{\beta'}$ invertible.

(b) By "the matrix acting on coordinates gives the coordinates of the image" (Lecture 4 notes, Theorem 2.2; Friedberg Theorem 2.14),

$$[\boldsymbol{v}]_\beta = [I_V(\boldsymbol{v})]_\beta = [I_V]^{\beta}_{\beta'}[\boldsymbol{v}]_{\beta'} = Q[\boldsymbol{v}]_{\beta'}.$$ □

Remark 1.1 (Three sentences about Q)**.**

- **What the columns are**: the j-th column of Q is $[\boldsymbol{b}'_j]_\beta$, that is, "what the j-th new basis vector looks like in the old basis". So Q is never found by solving equations: expand each vector of β' in β.
- **Which side to multiply on**: $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$ turns β'-coordinates into β-coordinates by **left** multiplication by Q; conversely $[\boldsymbol{v}]_{\beta'} = Q^{-1}[\boldsymbol{v}]_\beta$. To remember the direction, read $Q = [I_V]^{\beta}_{\beta'}$: subscript in, superscript out.
- **Two special cases**: if $\beta' = \beta$ then $Q = I_n$; if $V = \mathbb{F}^n$ and β is the standard basis, the columns of Q are simply the vectors of β' themselves, because coordinates in the standard basis are the entries.

Example 1.1 (Computing Q; Friedberg Example 1, §2.5, p. 112)**.** In $\mathbb{R}^2$ take $\beta = \{(1,1), (1,-1)\}$ and $\beta' = \{(2,4), (3,1)\}$. Find the matrix Q that changes β'-coordinates into β-coordinates, and compute $[(2,4)]_\beta$.

Approach. The columns of Q are the coordinate vectors of the vectors of β' in the basis β, so each of $(2,4)$ and $(3,1)$ has to be written as a combination of $(1,1)$ and $(1,-1)$; these are two small linear systems, and with these numbers the sum and the difference give the answer at once.

From

$$(2,4) = 3(1,1) - 1(1,-1), \qquad (3,1) = 2(1,1) + 1(1,-1)$$

we get

$$Q = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}, \qquad [(2,4)]_\beta = Q \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

The second identity is checked directly: $3(1,1) - 1(1,-1) = (2,4)$.

Theorem 1.4 (Changing the basis of an operator; Friedberg Theorem 2.23, §2.5, p. 112)**.** *Let T be a linear operator on a finite-dimensional space V, let β, β' be ordered bases of V, and let Q change β'-coordinates into β-coordinates. Then*

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q.$$

Proof. Write $I = I_V$. Since $T = IT = TI$, the matrix of a composition (Lecture 4 notes, Theorem 2.3; Friedberg Theorem 2.11) gives

$$Q[T]_{\beta'} = [I]^\beta_{\beta'}[T]^{\beta'}_{\beta'} = [IT]^\beta_{\beta'} = [TI]^\beta_{\beta'} = [T]^\beta_\beta [I]^\beta_{\beta'} = [T]_\beta Q,$$

and multiplying on the left by Q^{-1} finishes the proof. □

Proposition 1.5 (The matrix version; Friedberg §2.5, p. 115, Corollary)**.** *Let $A \in M_{n\times n}(\mathbb{F})$, let γ be an ordered basis of $\mathbb{F}^n$, and let Q be the matrix whose j-th column is the j-th vector of γ. Then*

$$[L_A]_\gamma = Q^{-1}AQ.$$

Proof. Take $T = L_A$ and β the standard basis in Theorem 1.4: then $[T]_\beta = A$, and $Q = [I]^\beta_{\beta'}$ has the vectors of γ as its columns (the third special case in Remark 1.1), so $[L_A]_\gamma = Q^{-1}AQ$. □

Definition 1.3 (Similar matrices; Friedberg §2.5, p. 115)**.** Let $A, B \in M_{n\times n}(\mathbb{F})$. If $B = Q^{-1}AQ$ for some invertible Q, then B is **similar** to A. By Theorem 1.4, the matrices of one operator in two bases are always similar.

Remark 1.2 (Similarity is an equivalence relation)**.** $A = IAI^{-1}$ gives reflexivity; $B = Q^{-1}AQ$ implies $A = QBQ^{-1}$, symmetry; and if $B = Q^{-1}AQ$ and $C = P^{-1}BP$, then $C = (QP)^{-1}A(QP)$, transitivity. So similarity classifies operators: the matrices in one class are the same operator in different clothes.

Example 1.2 (Changing the basis in practice; Friedberg Example 2, §2.5, p. 113)**.** Let T be the operator on $\mathbb{R}^2$ defined by $T(a,b) = (3a - b,\ a + 3b)$, with β, β' as in Example 1.1. Compute $[T]_{\beta'}$ and check it against the images of the basis vectors.

Approach. Take both roads. One is $Q^{-1}[T]_\beta Q$ from Theorem 1.4; the other goes back to the definition and expands $T(\boldsymbol{b}'_j)$ in β', to see whether the j-th column comes out the same. The first road is fast; the second is the only way to be sure the matrix is right.

In the standard basis $[T]_\beta = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$ (the first column is $T(1,0) = (3,1)$, the second is $T(0,1) = (-1,3)$), and Example 1.1 gives $Q = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ with $Q^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$. Hence

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ -2 & 2 \end{bmatrix}.$$

Check: $T(3,1) = (8,6) = 1\cdot(2,4) + 2\cdot(3,1)$, whose coefficients $(1,2)^\top$ are the second column (the second vector of β' is $(3,1)$); and $T(2,4) = (2,14) = -2\cdot(2,4) + 2\cdot(3,1)$, whose coefficients $(-2,2)^\top$ are the first column. Both columns match.

Remark 1.3 (What to practise)**.**

- Finding a change of coordinate matrix is nothing but "expand each new basis vector in the old basis"; that is the whole content of Lay §4.7 (p. 239–242, Exercises 1–9) and Friedberg §2.5 (p. 116, Exercises 3, 6).

- What sits on which side in Theorem 1.4 is the usual slip: $[T]_{\beta'}$ needs *both* ends in the new basis, hence Q on the left and Q^{-1} on the right, that is $Q^{-1}[T]_{\beta}Q$, not the other way round.
- Two drills worth doing: given A and γ, compute $[L_A]_\gamma$ (Friedberg Exercise 6, p. 116); and, to decide whether two matrices are similar, compare rank and invertibility first (see Section 3).

2 Dual Spaces

Push the idea of a matrix representation to its limit and ask: **can all linear transformations on V be packed into one vector space?** Yes — into V^*. Duality is "measuring vectors with linear functionals", and the rank computations of the next section use exactly this measuring tool.

Definition 2.1 (Linear functional; Friedberg §2.6, p. 119)**.** Let V be a vector space over $\mathbb{F}$. A linear map $f : V \to \mathbb{F}$ (whose codomain is the one-dimensional space $\mathbb{F}$) is a **linear functional** on V.

Example 2.1 (Three linear functionals to remember; Friedberg Examples 1–3, §2.6, p. 119)**.**

- **Integral type**: on $V = C[0, 2\pi]$ fix g and set $h(\boldsymbol{x}) = \frac{1}{2\pi}\int_0^{2\pi} x(t)g(t)\,dt$. With $g = \sin nt$ or $\cos nt$, $h(\boldsymbol{x})$ is the n-th Fourier coefficient.
- **Trace type**: on $V = M_{n\times n}(\mathbb{F})$, $f(A) = \operatorname{tr}(A)$.
- **Coordinate type**: fix an ordered basis $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ of V and let $f_i(\boldsymbol{x}) = a_i$, where $[\boldsymbol{x}]_\beta = (a_1, \dots, a_n)^\top$. The map f_i is the i-th **coordinate function**; the point is that $f_i(\boldsymbol{x}_j) = \delta_{ij}$.

Definition 2.2 (Dual space; Friedberg §2.6, p. 119)**.** The space $\mathcal{L}(V, \mathbb{F})$ of all linear functionals on V is the **dual space** of V, written V^*. If $\dim V = n$, then $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$ (Lecture 4 notes, Proposition 4.8) gives

$$\dim V^* = \dim V.$$

Theorem 2.1 (The dual basis; Friedberg Theorem 2.24, §2.6, p. 120)**.** *Let V be n-dimensional with ordered basis $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$, let f_i be the i-th coordinate function and $\beta^* = \{f_1, \dots, f_n\}$. Then β^* is an ordered basis of V^*, and for every $f \in V^*$,*

$$f = \sum_{i=1}^{n} f(\boldsymbol{x}_i)\, f_i.$$

Proof. First show that $g = \sum_i f(\boldsymbol{x}_i) f_i$ equals f: for each basis vector $\boldsymbol{x}_j$,

$$g(\boldsymbol{x}_j) = \sum_{i=1}^{n} f(\boldsymbol{x}_i) f_i(\boldsymbol{x}_j) = \sum_{i=1}^{n} f(\boldsymbol{x}_i)\delta_{ij} = f(\boldsymbol{x}_j),$$

and two linear functionals that agree on a basis are equal (the corollary to Lecture 4 notes, Theorem 4.1). So β^* spans V^*; since $|\beta^*| = n = \dim V^*$, it is a basis. □

Definition 2.3 (Dual basis; Friedberg §2.6, p. 120)**.** The ordered basis $\beta^* = \{f_1, \dots, f_n\}$ of Theorem 2.1 is the **dual basis** of β; its defining property is $f_i(\boldsymbol{x}_j) = \delta_{ij}$.

Example 2.2 (Computing a dual basis; Friedberg Example 4, §2.6, p. 120)**.** Let $\beta = \{(2, 1), (3, 1)\}$ be a basis of $\mathbb{R}^2$ and find the dual basis f_1, f_2.

Approach. The conditions $f_i(\boldsymbol{b}_j) = \delta_{ij}$ are four equations whose unknowns are the values of f_i on the standard basis. Writing $\boldsymbol{b}_j$ in terms of $\boldsymbol{e}_1, \boldsymbol{e}_2$ turns them into familiar linear systems; alternatively, read the answer off directly: f_1 must take the value 1 at $(2, 1)$ and 0 at $(3, 1)$.

For f_1, write $f_1(x,y) = ax + by$; then

$$2a + b = 1, \qquad 3a + b = 0,$$

so $a = -1$, $b = 3$, i.e. $f_1(x,y) = -x + 3y$; similarly $f_2(x,y) = x - 2y$. Check: $f_1(2,1) = 1$, $f_1(3,1) = 0$, $f_2(2,1) = 0$, $f_2(3,1) = 1$.

Theorem 2.2 (Transpose; Friedberg Theorem 2.25, §2.6, p. 121)**.** *Let V, W be finite-dimensional with bases β, γ, and let $T : V \to W$ be linear. Define $T^t : W^* \to V^*$ by*

$$T^t(g) = g \circ T, \qquad \textit{that is,} \qquad T^t(g)(\boldsymbol{x}) = g(T(\boldsymbol{x})).$$

Then T^t is linear and

$$[T^t]_{\gamma^*}^{\beta^*} = \big([T]_\beta^\gamma\big)^\top.$$

Proof. The linearity of T^t follows from the compatibility of composition with addition and scalar multiplication. Let $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$, $\gamma = \{\boldsymbol{y}_1, \dots, \boldsymbol{y}_m\}$ and $A = [T]_\beta^\gamma$, so that $T(\boldsymbol{x}_i) = \sum_k A_{ki}\boldsymbol{y}_k$. Expanding $T^t(g_j)$ in β^* by Theorem 2.1, its i-th coefficient is

$$(T^t(g_j))(\boldsymbol{x}_i) = g_j(T(\boldsymbol{x}_i)) = \sum_k A_{ki} g_j(\boldsymbol{y}_k) = A_{ji},$$

because $g_j(\boldsymbol{y}_k) = \delta_{jk}$; this is the entry in row i, column j of $A^\top$. □

Remark 2.1 (How to remember it)**.** Theorem 2.2 says: **the matrix of the transpose of a transformation is the transpose of its matrix**, with each side taken in the corresponding dual bases. Its use is to translate "left or right multiplication" into "transpose": the map $g \mapsto g \circ T$ and the map $A \mapsto A^\top$ are the same thing.

Example 2.3 (A transpose computed; Friedberg Example 5, §2.6, p. 122)**.** Let $T : P_1(\mathbb{R}) \to \mathbb{R}^2$ be $T(p(x)) = (p(0), p(2))$ with β, γ the standard bases. Then $[T]_\beta^\gamma = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$ and

$$[T^t]_{\gamma^*}^{\beta^*} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \big([T]_\beta^\gamma\big)^\top.$$

The check: expand $T^t(g_1)$ in $\beta^* = \{f_1, f_2\}$; its coefficients are determined by $T^t(g_1)(1)$ and $T^t(g_1)(x)$, and they agree with the columns of the matrix.

Lemma 2.3 (The lemma behind the double dual; Friedberg §2.6, p. 122)**.** *Let V be finite-dimensional and $\boldsymbol{x} \in V$. If $\hat{\boldsymbol{x}}(f) := f(\boldsymbol{x}) = 0$ for every $f \in V^*$, then $\boldsymbol{x} = \boldsymbol{0}$.*

Proof. If $\boldsymbol{x} \neq \boldsymbol{0}$, extend it to an ordered basis $\beta = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ with $\boldsymbol{x}_1 = \boldsymbol{x}$ and take f_1 from the dual basis; then $\hat{\boldsymbol{x}}(f_1) = f_1(\boldsymbol{x}) = 1 \neq 0$. □

Theorem 2.4 (The double dual; Friedberg Theorem 2.26, §2.6, p. 122)**.** *Let V be finite-dimensional and define $\psi : V \to V^{**}$ by $\psi(\boldsymbol{x}) = \hat{\boldsymbol{x}}$, where $\hat{\boldsymbol{x}}(f) = f(\boldsymbol{x})$. Then ψ is an isomorphism.*

Proof. Linearity: $f(\boldsymbol{x} + \boldsymbol{y}) = f(\boldsymbol{x}) + f(\boldsymbol{y})$ for every f, hence $\widehat{\boldsymbol{x} + \boldsymbol{y}} = \hat{\boldsymbol{x}} + \hat{\boldsymbol{y}}$, and similarly for scalar multiples. Injectivity: Lemma 2.3. So ψ is injective, and $\dim V^{**} = \dim V^* = \dim V$ makes it surjective as well. □

Proposition 2.5 (The converse question for dual bases; Friedberg §2.6, p. 123, Corollary)**.** *Let V be finite-dimensional. Then every ordered basis of V^* is the dual basis of some basis of V.*

Proof. Let $\{f_1, \dots, f_n\}$ be an ordered basis of V^* and regard it as a list of vectors of V^{**}. By Theorem 2.4 there are $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$ with $f_j = \hat{\boldsymbol{x}_j}$, that is $f_j(\boldsymbol{x}_i) = \delta_{ij}$. So $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ and $\{f_j\}$ are dual to each other. □

Remark 2.2 (Natural and not natural)**.** The assignment $\beta \mapsto \beta^*$ depends on β: change the basis and the whole dual basis changes (hence "V is isomorphic to V^*" is true, but no isomorphism is independent of a basis). The map ψ of Theorem 2.4, by contrast, **uses no basis at all** — it is the "natural" isomorphism, which is why we identify V with V^{**} but never with V^*.

Remark 2.3 (What to practise)**.**

- Computing a dual basis (Friedberg §2.6, p. 123, Exercises 3–5) is solving n linear equations in n unknowns; δ_{ij} is the only thing to remember.
- Both directions of the transpose have to be practised: given T, compute T^t (Exercise 7), and check it with $[T^t]_{\gamma^*}^{\beta^*} = ([T]_\beta^\gamma)^\top$.
- The most useful items in the true/false lists are "$\dim V^* = \dim V$" and Proposition 2.5; together they settle "is this basis a dual basis?" in one line.

3 The Rank of a Matrix

The last part of the lecture is **one number**: $\operatorname{rank}(A)$. It is "how many dimensions the image of A has", it is "how many columns of A are really independent", and it is "how many pivots a row reduction leaves". The task of this section is to prove that these three sentences say the same thing, and to see how the number behaves under a change of basis and under multiplication.

Definition 3.1 (Elementary operations and elementary matrices; Friedberg §3.1, p. 148–150)**.** An **elementary row operation** on a matrix is one of: interchanging two rows; multiplying a row by a nonzero scalar; adding a multiple of one row to another. Column operations are defined the same way. An $m \times m$ matrix obtained from I_m by one elementary row operation is an **elementary matrix**.

Theorem 3.1 (Elementary matrices realise elementary operations; Friedberg Theorem 3.1, §3.1, p. 149)**.** *Let $A \in M_{m\times n}(\mathbb{F})$ and suppose B is obtained from A by one elementary row (column) operation. Then there is an $m \times m$ ($n \times n$) elementary matrix E with*

$$B = EA \qquad (\textit{respectively } B = AE),$$

where E is obtained from I_m (I_n) by the same operation.

Theorem 3.2 (Elementary matrices are invertible; Friedberg Theorem 3.2, §3.1, p. 150)**.** *Every elementary matrix is invertible, and its inverse is an elementary matrix of the same type.*

Proof. Let E come from I_n by one elementary row operation. Undo that step (interchange again; multiply by $1/c$; subtract the same multiple), and E goes back to I_n by an operation of the same type. By Theorem 3.1 there is an elementary matrix E' of that type with $E'E = I_n$, so E is invertible and $E^{-1} = E'$. □

Definition 3.2 (Rank; Friedberg §3.2, p. 152; Lay §4.6, p. 233)**.** Let $A \in M_{m\times n}(\mathbb{F})$ and let $L_A : \mathbb{F}^n \to \mathbb{F}^m$ be its left-multiplication map. Define

$$\operatorname{rank}(A) = \dim \mathcal{R}(L_A) = \dim \operatorname{Col}(A),$$

the dimension of the column space of A (the span of its columns). Then the dimension theorem (Lecture 3 notes) gives $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$.

Theorem 3.3 (Rank does not depend on the representation; Friedberg Theorem 3.3, §3.2, p. 152)**.** *Let $T : V \to W$ be linear and let β, γ be ordered bases of V, W. Then*

$$\operatorname{rank}(T) = \operatorname{rank}\big([T]_\beta^\gamma\big).$$

Proof. By Lecture 4 notes, Theorem 2.2, the maps $\boldsymbol{v} \mapsto [T(\boldsymbol{v})]_\gamma$ and $[\boldsymbol{v}]_\beta \mapsto [T]_\beta^\gamma [\boldsymbol{v}]_\beta$ are the same assignment, and the coordinate mappings are isomorphisms (Theorem 1.2), so the two images have the same dimension. □

Theorem 3.4 (Multiplying by invertible matrices preserves the rank; Friedberg Theorem 3.4, §3.2, p. 153)**.** *Let A be $m \times n$ and let P, Q be invertible of sizes $m \times m$ and $n \times n$. Then*

$$\operatorname{rank}(AQ) = \operatorname{rank}(PA) = \operatorname{rank}(PAQ) = \operatorname{rank}(A).$$

In particular (with P, Q products of elementary matrices): ***elementary row and column operations preserve the rank.***

Proof. L_Q is onto (as Q is invertible), so $\mathcal{R}(L_{AQ}) = L_A(L_Q(\mathbb{F}^n)) = L_A(\mathbb{F}^n) = \mathcal{R}(L_A)$, and taking dimensions gives $\operatorname{rank}(AQ) = \operatorname{rank}(A)$. Also $\operatorname{rank}(PA) = \operatorname{rank}((PA)^\top) = \operatorname{rank}(A^\top P^\top)$ with $P^\top$ invertible, so the result just proved applies (using $\operatorname{rank}(A) = \operatorname{rank}(A^\top)$, see Proposition 3.6). The two identities together give $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$. □

Theorem 3.5 (Rank is the maximal number of independent columns; Friedberg Theorem 3.5, §3.2, p. 153)**.** *The rank of a matrix equals the maximal number of linearly independent columns, that is, the dimension of the subspace spanned by its columns.*

Proof. $\operatorname{rank}(A) = \dim \mathcal{R}(L_A)$, and $\mathcal{R}(L_A) = \operatorname{Span}\{A\boldsymbol{e}_1, \ldots, A\boldsymbol{e}_n\} = \operatorname{Span}\{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\}$ (Lecture 4 notes, Theorem 2.1(b)), which is exactly the column space. □

Proposition 3.6 (Row rank equals column rank; Friedberg §3.2, p. 158, Corollary 2)**.** *For every $m \times n$ matrix A,*

$$\operatorname{rank}(A^\top) = \operatorname{rank}(A),$$

and $\operatorname{rank}(A)$ *is also the maximal number of linearly independent rows; the row space and the column space have the same dimension.*

Theorem 3.7 (The normal form for a given rank; Friedberg Theorem 3.6, §3.2, p. 155)**.** *Let A be $m \times n$ of rank r. Then $r \le m$, $r \le n$, and finitely many elementary row and column operations transform A into*

$$D = \begin{bmatrix} I_r & O_1 \\ O_2 & O_3 \end{bmatrix},$$

where O_1, O_2, O_3 are zero matrices (that is $D_{ii} = 1$ for $i \le r$ and all other entries 0).

Proposition 3.8 (Two consequences of the normal form; Friedberg §3.2, p. 158–159)**.**

(a) A of rank r can be written as $D = BAC$ with B, C invertible (Corollary 1);
(b) every invertible matrix is a product of elementary matrices (Corollary 3).

Proof. (a) Every elementary operation is a left or right multiplication by an elementary matrix (Theorem 3.1); multiply those matrices in order to get B, C, which are invertible by Theorem 3.2 and by "a product of invertible matrices is invertible".

(b) If A is invertible then $\operatorname{rank}(A) = n$, so the normal form is $D = I_n$; by (a), $I_n = BAC$, hence $A = B^{-1}C^{-1}$, and B^{-1}, C^{-1} are products of elementary matrices. □

Theorem 3.9 (The Rank Theorem; Lay Theorem 14, §4.6, p. 233)**.** *Let A be $m \times n$. Then the column space and the row space have the same dimension, this common dimension is* $\operatorname{rank}(A)$*, it equals the number of pivot positions of A, and*

$$\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n.$$

Proof. The identity $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$ is the dimension theorem applied to $L_A : \mathbb{F}^n \to \mathbb{F}^m$ (Lecture 3 notes). For "$\operatorname{rank}(A)$ = the number of pivots": row reduction preserves the rank (Theorem 3.4), while in echelon form the pivot columns are linearly independent and every other column is a combination of them, so the rank is the number of pivots (Theorem 3.5). Row rank equals column rank is Proposition 3.6. □

Theorem 3.10 (Monotonicity of the rank; Friedberg Theorem 3.7, §3.2, p. 159)**.** *Let $T : V \to W$, $U : W \to Z$ be linear and let A, B be matrices for which AB is defined. Then*

$$\operatorname{rank}(UT) \le \operatorname{rank}(U), \qquad \operatorname{rank}(UT) \le \operatorname{rank}(T),$$

$$\operatorname{rank}(AB) \le \operatorname{rank}(A), \qquad \operatorname{rank}(AB) \le \operatorname{rank}(B).$$

Proof. Because

$$\mathcal{R}(UT) = U\big(\mathcal{R}(T)\big) \subseteq \mathcal{R}(U),$$

taking dimensions gives the first inequality. (b): put $A' = [U]_\gamma^\beta$ and $B' = [T]_\alpha^\beta$; by Lecture 4 notes, Theorem 2.3, $[UT]_\alpha^\gamma = A'B'$, hence by Theorem 3.3

$$\operatorname{rank}(UT) = \operatorname{rank}(A'B') \le \operatorname{rank}(B') = \operatorname{rank}(T).$$

(c)(d) are the matrix versions of (a)(b); for (d) use the transpose, $\operatorname{rank}(AB) = \operatorname{rank}(B^\top A^\top) \le \operatorname{rank}(B^\top) = \operatorname{rank}(B)$. □

Remark 3.1 (The Invertible Matrix Theorem in rank language; Lay §4.6, p. 235)**.** For an $n \times n$ matrix A, each of the following is equivalent to "A is invertible": the columns of A form a basis of $\mathbb{R}^n$; $\operatorname{Col}(A) = \mathbb{R}^n$; $\dim \operatorname{Col}(A) = n$; $\operatorname{rank}(A) = n$; $\operatorname{Nul}(A) = \{\mathbf{0}\}$; $\dim \operatorname{Nul}(A) = 0$. They continue the Invertible Matrix Theorem of Lecture 4, merely replacing the language of "$A\boldsymbol{x} = \boldsymbol{b}$" by the language of rank.

Remark 3.2 (What to practise)**.**

- The fixed drill for a rank: row-reduce to echelon form and count the nonzero rows (equivalently the pivots). Lay §4.6 (p. 230–236, Exercises 1–23) and Friedberg §3.2 (p. 165, Exercises 2, 3, 9) are all this drill.
- Theorem 3.4 is the hinge of the section: similar matrices have equal rank, so the rank is an invariant of the operator while the entries are not.
- Use the rank inequalities to trap a quantity between bounds: e.g. to show that a composite cannot increase the dimension (Theorem 3.10), or to decide whether a given rank is possible.

4 A Worked Example

Example 4.1 (Change of basis, dual basis and rank in one pass; written for this lecture; the methods are those of Friedberg §2.5, §2.6, §3.2 and Lay §4.6, §4.7)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have standard matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 0 \end{bmatrix},$$

let $\beta' = \{(1,0,0),(1,1,0),(1,1,1)\}$ be an ordered basis of $\mathbb{R}^3$, and let β be the standard basis. (1) Write the change of coordinate matrix $Q = [I]^{\beta}_{\beta'}$ and find Q^{-1}; (2) compute $[T]_{\beta'}$ as $Q^{-1}AQ$ and check it against the images of the $\boldsymbol{b}'_j$; (3) find $\operatorname{rank}(A)$ and $\dim \operatorname{Nul}(A)$, and explain why $\operatorname{rank}([T]_{\beta'}) = \operatorname{rank}(A)$; (4) find the dual basis f_1, f_2, f_3 of β' and verify $f_i(\boldsymbol{b}'_j) = \delta_{ij}$; (5) let B be the elementary matrix obtained by interchanging rows 1 and 2 of I_3, and verify that $\operatorname{rank}(BA) = \operatorname{rank}(A)$.

Approach. Each part matches one theorem. (1) is "the columns of Q are the new basis vectors in the old basis"; here the old basis is the standard one, so the columns of Q are the three vectors themselves. (2) is Theorem 1.4. (3) is "rank = number of pivots" plus Theorem 3.3. (4) is "the dual basis is fixed by $f_i(\boldsymbol{b}'_j) = \delta_{ij}$", read off from the formula that solves for β'-coordinates. (5) is Theorem 3.4: elementary matrices are invertible, so left multiplication by one cannot change the rank. The skeleton of the whole example is one sentence: *a change of basis changes the look of the matrix, never its rank.*

(1) The change of coordinate matrix. Since β is the standard basis, the columns of Q are the three vectors of β':

$$Q = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad Q^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

(Q is upper triangular with all diagonal entries 1, so its inverse is read off immediately: $\boldsymbol{e}_1 = \boldsymbol{b}'_1$, $\boldsymbol{e}_2 = \boldsymbol{b}'_2 - \boldsymbol{b}'_1$, $\boldsymbol{e}_3 = \boldsymbol{b}'_3 - \boldsymbol{b}'_2$, and these coefficients, taken as columns, give Q^{-1}.)

(2) The matrix in the new basis. First

$$AQ = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 0 & 1 & 1 \end{bmatrix},$$

then multiply on the left by Q^{-1} (subtract the next row from each row):

$$[T]_{\beta'} = Q^{-1}AQ = \begin{bmatrix} -1 & -3 & -4 \\ 2 & 5 & 7 \\ 0 & 1 & 1 \end{bmatrix}.$$

Check: $T(\boldsymbol{b}'_1) = A(1,0,0)^{\top} = (1,2,0)^{\top} = -1\cdot\boldsymbol{b}'_1 + 2\cdot\boldsymbol{b}'_2$, whose coefficients $(-1,2,0)^{\top}$ are the first column; $T(\boldsymbol{b}'_2) = (3,6,1)^{\top} = -3\boldsymbol{b}'_1 + 5\boldsymbol{b}'_2 + 1\boldsymbol{b}'_3$, whose coefficients $(-3,5,1)^{\top}$ are the second column; and $T(\boldsymbol{b}'_3) = (4,8,1)^{\top} = -4\boldsymbol{b}'_1 + 7\boldsymbol{b}'_2 + \boldsymbol{b}'_3$, whose coefficients $(-4,7,1)^{\top}$ are the third column. All three match.

(3) Rank and nullity. Row-reduce A: subtract twice the first row from the second,

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so there are two pivots and $\operatorname{rank}(A) = 2$; the dimension theorem then gives $\dim \operatorname{Nul}(A) = 3-2 = 1$. Indeed, solving $A\boldsymbol{x} = \boldsymbol{0}$ gives $x_2 = 0$ and $x_1 + x_3 = 0$, that is

$$\operatorname{Nul}(A) = \operatorname{Span}\{(-1,0,1)^{\top}\}.$$

For $[T]_{\beta'}$ the same thing happens: subtracting rows produces a zero row (the third row is a combination of the first two), so $\operatorname{rank}([T]_{\beta'}) = 2 = \operatorname{rank}(A)$, in agreement with Theorem 3.3.

(4) The dual basis. Write $\boldsymbol{x} = (x_1, x_2, x_3)$ in the basis β': $\boldsymbol{x} = c_1\boldsymbol{b}'_1 + c_2\boldsymbol{b}'_2 + c_3\boldsymbol{b}'_3$, and comparing entries gives $c_3 = x_3$, $c_2 = x_2 - x_3$, $c_1 = x_1 - x_2$. The coordinate functions are the dual basis:

$$f_1(\boldsymbol{x}) = x_1 - x_2, \qquad f_2(\boldsymbol{x}) = x_2 - x_3, \qquad f_3(\boldsymbol{x}) = x_3.$$

Verification: $f_1(\boldsymbol{b}'_1) = 1$, $f_1(\boldsymbol{b}'_2) = 0$, $f_1(\boldsymbol{b}'_3) = 0$; $f_2(\boldsymbol{b}'_1) = 0$, $f_2(\boldsymbol{b}'_2) = 1$, $f_2(\boldsymbol{b}'_3) = 0$; $f_3(\boldsymbol{b}'_1) = f_3(\boldsymbol{b}'_2) = 0$, $f_3(\boldsymbol{b}'_3) = 1$. That is $f_i(\boldsymbol{b}'_j) = \delta_{ij}$.

(5) Elementary operations do not change the rank. B is obtained from I_3 by interchanging the first two rows, hence is invertible (Theorem 3.2), and Theorem 3.4 gives $\operatorname{rank}(BA) = \operatorname{rank}(A) = 2$. Directly:

$$BA = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

whose first row is twice the second while the remaining two rows are not proportional, so the rank is still 2.

Remark 4.1 (Not covered here — these belong to later lectures)**.**

- **Partitioned matrices and LU factorizations**: Lay §2.4, p. 123, §2.5, p. 132; Friedberg near §3.3–§3.4. Here elementary matrices only implement row operations; matrices are not yet split into a lower times an upper triangular factor.
- **Determinants**: Lecture 7 (Lay §3.1, p. 164; Friedberg §4.1). This lecture avoids them on purpose: similarity, rank and invertibility can all be discussed without them.
- **Eigenvalues and diagonalization**: Lecture 9 (Lay §5.2–§5.3; Friedberg §5.1–§5.2). Lecture 9 returns to similarity and asks which Q diagonalises $[T]_\beta$.
- **Inner products and adjoints**: Lecture 11 (Lay §6.1; Friedberg §6.1–§6.2). The geometric version of duality appears there.

5 Checklist and Exercises

By the end of this lecture you should be able to:

1. state the definitions of the coordinate vector and the coordinate mapping, and explain why the coordinate mapping is an isomorphism (uniqueness gives linearity, spanning gives surjectivity, independence gives injectivity);
2. write down the change of coordinate matrix $Q = [I_V]^{\beta}_{\beta'}$, say that its columns are the vectors of β' in β, and state $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$ on the spot;
3. change the basis of a 2×2 operator with $Q^{-1}[T]_\beta Q$ and check the result against the images of the basis vectors;
4. define similar matrices, prove that similarity is an equivalence relation, and say that rank and invertibility are invariant under similarity;
5. give the three examples of linear functionals (integral, trace, coordinate) and explain why $\dim V^* = \dim V$;
6. compute the dual basis of a given basis by solving the δ_{ij} equations, and say precisely what "the two bases are dual to each other" means;
7. state $T^t(g) = g \circ T$ and $[T^t]^{\beta^*}_{\gamma^*} = ([T]^{\gamma}_{\beta})^{\top}$, and explain why "the matrix of the transpose is the transpose of the matrix";
8. state the double dual theorem and explain why ψ is natural while an isomorphism $V \cong V^*$ is not;
9. compute the rank of a matrix by counting pivots in a row reduction, or by counting a maximal independent set of columns, and pass between the two descriptions;
10. say that rank is preserved by multiplication by invertible matrices and by elementary operations, and state the Rank Theorem $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$;
11. use the rank inequalities (Theorem 3.10) to bound the rank of a composite or a product from above and below.

Exercises (printed pages). Lay: §4.4 p. 222 (Exercises 1–13: coordinate vectors and the coordinate mapping), §4.7 p. 242 (Exercises 1–9: change-of-coordinates matrices), §4.6 p. 236

(Exercises 1–23: rank, row space, the Rank Theorem), §2.2 p. 108 (Exercises 1–21: elementary matrices and inverses), §2.3 p. 114 (exercises on the Invertible Matrix Theorem); Friedberg: §2.5 p. 116 (Exercises 2, 3, 6, 8: change of basis and similarity), §2.6 p. 123 (Exercises 2–7: dual bases and the transpose), §3.1 p. 151 (Exercises 1–8: elementary matrices), §3.2 p. 165 (Exercises 1–15: rank, normal form, rank inequalities). Each section also has Practice Problems; do those first.

How to combine the two texts:

- **Compute and prove are divided cleanly.** How to find a change of coordinate matrix, how to count a rank: Lay (§4.4, §4.7, §4.6). Why the change of basis is $Q^{-1}AQ$, why the double dual is natural, why the rank does not depend on the representation: Friedberg (§2.5, §2.6, §3.2).
- The core chain of the lecture: *the coordinate mapping is an isomorphism* $\to$ *the change of coordinate matrix* $Q = [I_V]^{\beta}_{\beta'}$ $\to$ *the operator changes basis as* $[T]_{\beta'} = Q^{-1}[T]_{\beta}Q$ $\to$ *similarity is "the operator in different clothes"* $\to$ *the rank survives the change of clothes.* The second thread is *duality*: *linear functionals* $\to$ *the dual basis* $\to$ *the transpose* $\to$ *the double dual is natural.*
- The single most important statement is Theorem 3.3: it welds "the rank of an operator" to "the rank of its matrix", so that the rank is at once a geometric quantity (the dimension of the image) and a computable one (the number of pivots), and Theorem 3.4 says it is also an **invariant**.